INFLECTION

Real-time multilingual sentiment analysis

MSML 641 - Final Project Presentation

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Problem & Users:

Persona

- Customer service agents at BPOs and international support orgs
- The QA managers and team leads above them
- Handling calls from customers not fluent in the agent's native language

Pain

- Agents can still carry the conversations with language barriers but can't distinguish tension building from the caller
- Tone is completely lost once words are translated
- Managers review emotionally neutral text after the call is over

What was done before?

- Forced to hire native speakers per language → costly and doesn't scale
- Enterprise platforms (SupportLogic & CallMiner) → priced for large scale call centers
- Generic translation with additional post-call reviews → emotion is lost in the transcription

The Gap:

Tools often sold for voice sentiment analysis only scan the transcript, and not the customer's voice. There is no tool in the space that combines real-time translation with acoustic sentiment analysis while remaining affordable for low to medium sized operations.

The Product:

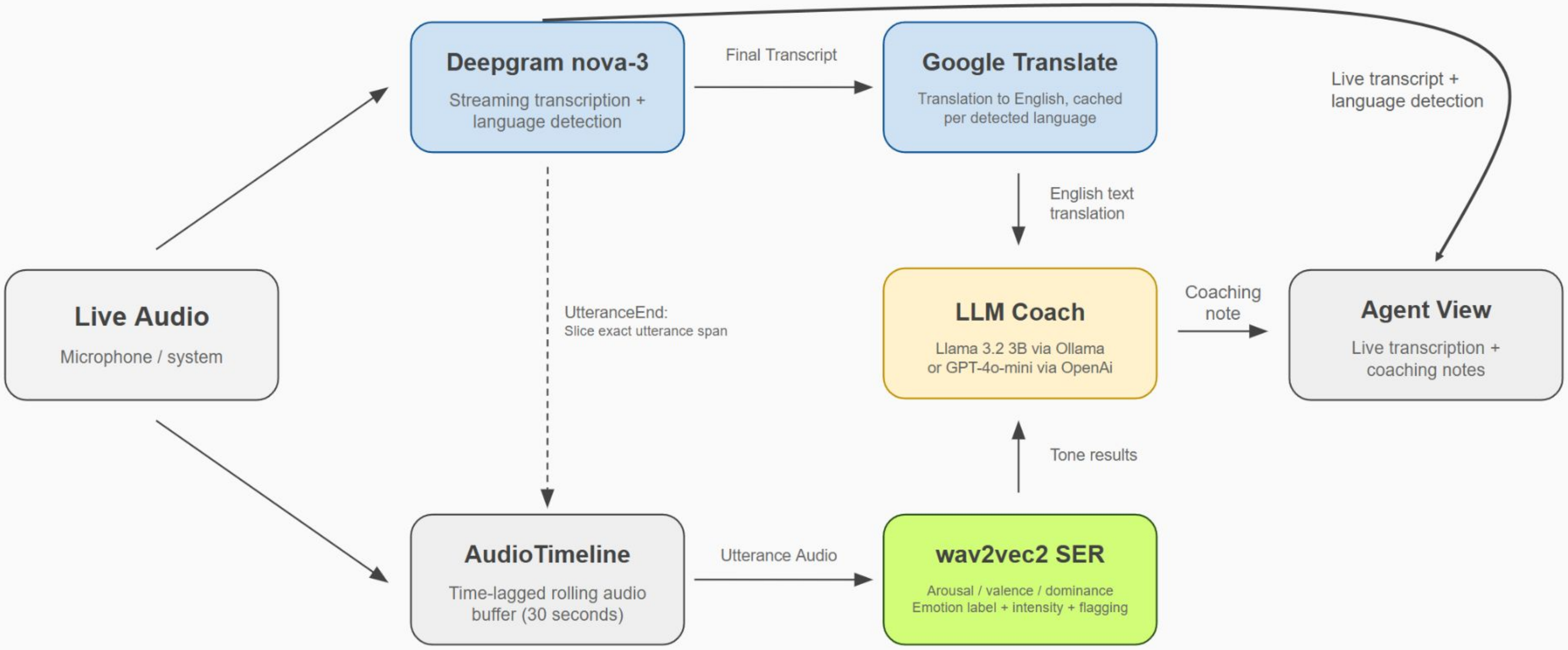
A standalone desktop companion. Inflection runs on its own, beside the agent's calling app (Zoom, phone call, etc.), and just listens to the audio output without any integrations or plugins

What the agent sees: there is a live transcript in the original language as well as an English translation, and the customer's emotional state read from their tone of voice. The LLM coach then converts the flags into actionable feedback for the agent.

The emotion model runs on-device; hosted components use free-tier options by default, with an optional OpenAI backend.

Original Product Design with Prototype UI. Current product operates in the terminal.

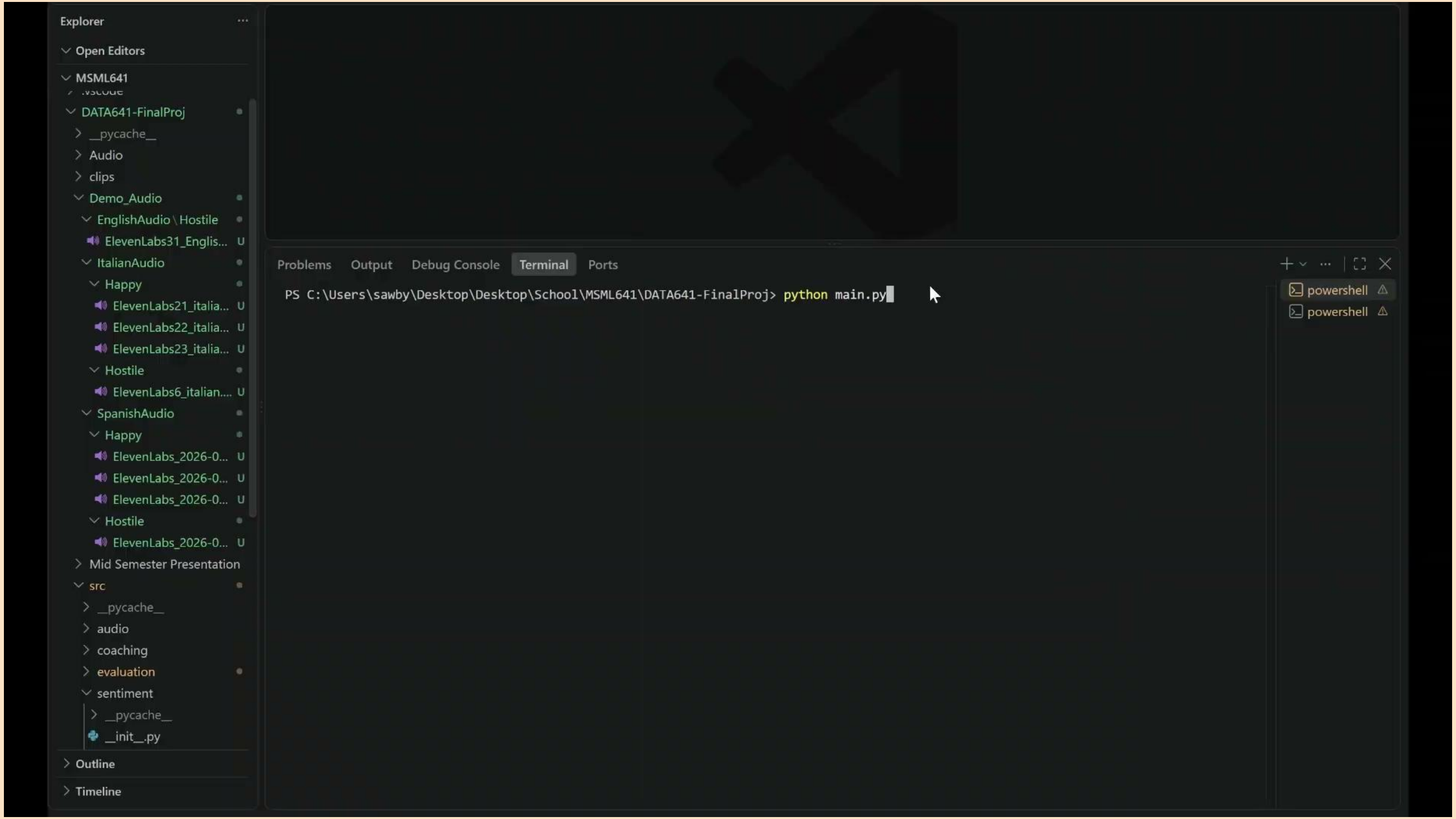
Architecture Diagram:



Blue = hosted API components || **Green** = Local acoustic model || **Yellow** = local LLM

Utterance Alignment: Every emotion reading covers exactly one complete statement → Transcript, translation, and emotion always describe the same audio

Demo


Explorer
Open Editors
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DATA641-FinalProj
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Mid Semester Presentation
src
__pycache__
audio
coaching
evaluation
sentiment
__pycache__
__init__.py
Outline
Timeline
Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\sawby\Desktop\Desktop\School\MSML641\DATA641-FinalProj> python main.py
powershell
powershell

[Video Link]

NLP Methods:

Speech Recognition

- ***Deepgram nova-3***
- Transcribes and detects the spoken language per utterance
- 10 available languages: English, Spanish, French, German, Hindi, Russian, Portuguese, Japanese, Italian, Dutch

Machine Translation

- ***Google Translate*** → Invoked per finalized utterance
- Translator instances cached per detected language
- Pipeline translates Non-English to English

Speech Emotion Recognition

- Pretrained ***wav2vec2*** → reads emotion from tone of voice
- Regression head outputs arousal / valence / dominance
- Rule-based mapping turns the three scores into a readable label, an intensity score, and an escalation flag

LLM Coaching

- ***Llama 3.2B***, running locally through ***Ollama*** (or ***GPT-4o-mini*** via ***OpenAI*** API)
- Prompted with the English-translated transcript and wav2vec2 tone results
- Generates a short, actionable coaching note for the supervisor when a hostile utterance is detected

Evaluation Design:

Data

- 90 pre-recorded utterances (30 sentences × 3 languages: EN/IT/ES)
- 3 categories: hostile, neutral, happy (10 each)
- Same emotional delivery across languages for consistency

Metric

- Tone-text agreement (manual rubric)
- Coaching trigger precision (automated)
- Processing latency (automated)
- Cross-language consistency (automated)

Baselines

- Ollama llama 3.2:3b (local, free)
- OpenAI gpt-4o-mini (API, paid)
- Same test set run on both for direct comparison

Results:

Tone-text agreement	Ollama	OpenAI
Tone alignment	2.57/3	2.83/3
Actionability	2.71/3	2.33/3

Trigger Precision	Ollama	OpenAI
Overall	75.56%	76.67%
False Positives	2	1
False Negatives	20	20

Latency	Ollama	OpenAI
Avg	4,579ms	1,630ms
Min	3,306ms	1,081ms
Max	11,094ms	3,437ms

Cross-Language Consistency	EN	IT	ES
Ollama	90%	10%	0%
OpenAI	90%	10%	0%

User Evidence

Who & How:

- Moderated sessions with native Italian and Japanese speakers.
- Five-minute guided conversations, one per target emotional state.
- Informal live testing with speakers in natural conversation.

What we Learned:

- Automatic language detection allowed for fewer additional configurations
- Transcript and Translations performed well during live use
- Wav2vec was mainly trained on english
- Tone was occasionally confused with which strong emotions it was hearing

What we Changed:

- Acoustic tone analysis reliably separated intense from neutral emotion, motivating its use as the primary escalation signal
- Updated the coach to read English translation after testing revealed that it was reading the original-language text

Ethics and Limitations:

Privacy & Consent:

- Live call audio is still personal data
- Deployments must adhere to call-recording and wiretap consent laws (which vary by jurisdiction)
- Audio and transcripts leave the device → Transcription process uses third-party infrastructure (Deepgram/Google)
- Emotion model and coach LLM stay local, limiting exposure

Bias:

- Emotion model was tuned on mostly English-language emotional speech
- Emotional expression can vary in different languages/cultures (specifically differences in volume, pace, etc.)
- Escalation flag firing more readily for some accents could encode bias into how customers are perceived

When the model is wrong:

- False flags can cause agents to overcorrect and managers to misjudge calls
- Missed flags create false confidence that a call is going better than it really is
- Mistranslation misinforms agents, poor coaching suggestions can worsen bad situations

Limitations:

- 10 available languages
- System capture is Windows-first
- Sentiment latency scales with utterance length

Thank you!